BUSINESS REQUIREMENTS DOCUMENT

Starbucks Beverage Analytics Dashboard

Power BI executive dashboard | Food & Beverage | Version 1.0

Document control	Detail
Project sponsor	Business Intelligence Portfolio
Prepared	August 2026
Status	Portfolio project requirements
Primary users	Business managers, marketing, product, nutrition, and BI teams

1. Purpose and Business Context

This document defines the business requirements for an interactive Power BI dashboard that consolidates Starbucks beverage nutrition and global store-presence information. The dashboard is intended to replace fragmented static review with a consistent, filterable executive experience.

Business problem

Stakeholders need to compare beverage categories, understand calories, sugar, caffeine, and protein attributes, identify high-caffeine drinks, and explore the recorded geographic footprint. Nutrition and store information may originate at different grains and must not be combined through an artificial relationship.

Business outcomes

Outcome	Success signal
Faster product comparison	Users can filter a category and view KPIs and rankings without manual spreadsheet work.
Consistent metrics	Cards, charts, and tooltips use governed DAX definitions.
Clear footprint view	Users can inspect presence by source-supported country or region.
Maintainable asset	Queries, model assumptions, and calculations are documented for handover.

In scope

• Beverage-level nutrition analysis
• Category comparisons and caffeine rankings
• Store-presence exploration at source grain
• Interactive Power BI visualisations, filters, and documentation

Out of scope

• Sales, margin, demand, clinical health outcomes, or consumer preference analysis unless new governed data is introduced
• Causal claims linking nutrition to store-network performance

2. Stakeholders and User Requirements

Stakeholder	Need	Representative question
Business manager	Portfolio and footprint overview	How is the visible beverage portfolio distributed by category?
Marketing team	Product comparison	Which categories warrant lower-sugar messaging exploration?
Product manager	Item-level detail	Which drinks lead the selected caffeine ranking?
Nutrition analyst	Transparent attributes	How do calories, sugar, caffeine, and protein compare at documented serving basis?
BI team	Governed and reusable model	Are metrics consistent under slicers and refresh?

User stories

ID	User story	Acceptance criterion
US-01	As a manager, I need headline KPIs so I can understand the active selection quickly.	Cards show beverage count and average calories, sugar, and caffeine with units.
US-02	As an analyst, I need category comparisons so I can identify patterns.	Category visuals cross-filter applicable nutrition visuals.
US-03	As a product user, I need a top-caffeine ranking.	Ranking respects selected category/filter context.
US-04	As a market user, I need a geographic view.	Store presence is aggregated only at the confirmed source grain.
US-05	As a report consumer, I need clear definitions.	Tooltips or documentation state units, serving basis, and metric logic.

3. Functional Requirements

ID	Requirement	Priority	Acceptance criteria
FR-01	Load the approved CSV/Excel sources through Power Query.	Must	Refresh completes without type errors; raw-to-clean row counts are recorded.
FR-02	Standardise category, beverage, and geography labels.	Must	Whitespace/casing rules are applied; exceptions are reviewable.
FR-03	Display total beverage count and average nutrition KPIs.	Must	Metrics update according to applicable filters and show units.
FR-04	Visualise beverage distribution by category.	Must	Counts/shares reflect the active category and report context.
FR-05	Visualise average calories, sugar, and caffeine by category.	Must	Visuals use consistent measure definitions and descriptive titles.
FR-06	Show top-ranked caffeine beverages.	Must	Top 5 logic honours selected filters and handles ties consistently.
FR-07	Show global store presence.	Must	Map/chart aggregation matches the documented source grain.
FR-08	Provide filters and reset path.	Should	Users can identify active selections and return to default state.
FR-09	Expose source/date and metric definitions.	Should	Report or documentation shows refresh/source version and definitions.

Non-functional requirements

Area	Requirement
Performance	Report opens and common interactions remain responsive on the intended Power BI Service capacity.
Accessibility	Use readable contrast, clear labels, meaningful titles, keyboard-friendly filters, and alternative text where supported.
Security	Apply workspace access and row-level security only after roles and geography ownership are defined.
Governance	Maintain source provenance, refresh logic, transformation steps, and a data dictionary.

4. Data, Metrics, and Design Requirements

Data requirements

Data area	Required fields / controls
Beverage nutrition	Stable beverage identifier or name; category; calories; sugar; caffeine; protein; documented serving basis and units.
Store presence	Country/region and either store identifier or pre-aggregated store count; reporting date and source grain.
Reference mapping	Approved category labels/sort order and standardised country/region mapping where needed.
Quality controls	Null, duplicate, zero, outlier, and type rules documented before publishing.

Metric requirements

Metric	Definition	Rule
Total Beverage Count	Distinct count of stable beverage key/name	Do not inflate from duplicated records.
Average Calories	Mean calories in current valid filter context	Confirm equivalent serving basis.
Average Sugar (g)	Mean valid sugar grams in current filter context	Display grams and blank handling.
Average Caffeine (mg)	Mean valid caffeine milligrams in current filter context	Display milligrams and decaf handling.
Top 5 Caffeine	Dense rank by caffeine within ALLSELECTED context	Retain user-selected category/filter context.

Dashboard requirements

The executive page must include KPI cards, category distribution, category nutrition comparisons, a top-caffeine visual, a store-presence visual, and relevant slicers. Visual titles must state the population and unit; tooltips must add detail without duplicating the page. Geography controls must not alter beverage metrics unless a valid shared model dimension is approved.

5. Acceptance, Risks, and Delivery Plan

Acceptance checklist

Area	Exit criterion
Data	Source timing, grain, units, and raw-to-clean record reconciliation are documented.
Model	Relationships have a clear business key; filter directions are tested; no ambiguous paths exist.
Measures	DAX outputs match manual sample checks under default and filtered states.
Visuals	Titles, units, tooltips, interactions, colours, and no-data states are reviewed.
Publishing	Workspace access, refresh approach, source date, and documentation are available to intended users.

Risks and mitigations

Risk	Mitigation
Different serving sizes make comparisons misleading.	Show serving basis and add a normalised measure only when source data supports it.
Store data is pre-aggregated and accidentally summed twice.	Confirm grain and aggregate once at the documented level.
Missing nutrition values distort averages.	Separate blanks from true zeroes and document inclusion rules.
Users infer sales or health outcomes from descriptive data.	State scope and limitations in the report and tooltips.

Delivery plan

1. Validate sources and data dictionary. 2. Build and audit Power Query transformations. 3. Model tables and create DAX measures. 4. Develop and test the executive report. 5. Conduct stakeholder review and accessibility QA. 6. Publish with documentation, refresh guidance, and governance controls.

Approval

Approval confirms that business definitions, source limitations, model grain, visual interactions, and acceptance criteria are understood by the project owner and intended stakeholders.